WORKED SOLUTION

# STEP II 2022, Question 3

Pure •

**STEP II 2022, Question 3**

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

## The key idea

The recurrence $F_{r+1} - F_r - F_{r-1} = 0$ is the whole engine. Summing it against powers of $\frac{1}{10}$ and re-indexing turns each piece into a multiple of $S_n$ plus a few boundary terms, which is exactly the generating-function trick. Part (i) is the safety net: it says the Fibonacci numbers grow no faster than powers of 2, so dividing by powers of 10 (or 100) gives tails that are tiny and easy to bound.

## Solution

### Part (i)

First, every $F_k \geq 0$ (by induction, since $F_0, F_1 \geq 0$ and sums of non-negative numbers are non-negative). Next, the sequence never decreases: $F_1 \geq F_0$, and for $k \geq 1$ we have $F_{k+1} - F_k = F_{k-1} \geq 0$.

So for every $k \geq 1$,

$$F_{k+1} = F_k + F_{k-1} \leq F_k + F_k = 2F_k.$$

Now fix $n \geq 1$ and induct on $r \geq n$. When $r = n$ the claim reads $F_n \leq F_n$. If $F_r \leq 2^{r-n}F_n$ for some $r \geq n$, then, since $r \geq 1$,

$$F_{r+1} \leq 2F_r \leq 2 \cdot 2^{r-n}F_n = 2^{(r+1)-n}F_n.$$

This completes the induction, so $F_r \leq 2^{r-n}F_n$ whenever $n \geq 1$ and $r \geq n$.

### Part (ii)

We rewrite each of the three sums on the left in terms of $S_n$.

*First sum.* Put $k = r + 1$:

$$\sum_{r=1}^{n} \frac{F_{r+1}}{10^{r-1}} = 100 \sum_{k=2}^{n+1} \frac{F_k}{10^k} = 100\Big(S_n - \frac{F_1}{10} + \frac{F_{n+1}}{10^{n+1}}\Big)$$
$$= 100S_n - 10F_1 + \frac{F_{n+1}}{10^{n-1}}.$$

*Second sum.* This is simply $10S_n$.

*Third sum.* Put $k = r - 1$:

$$\sum_{r=1}^{n} \frac{F_{r-1}}{10^{r-1}} = \sum_{k=0}^{n-1} \frac{F_k}{10^k} = F_0 + S_n - \frac{F_n}{10^n}.$$

Subtracting the second and third sums from the first gives

$$(100 - 10 - 1)S_n - 10F_1 - F_0 + \frac{F_n}{10^n} + \frac{F_{n+1}}{10^{n-1}} = 89S_n - 10F_1 - F_0 + \frac{F_n}{10^n} + \frac{F_{n+1}}{10^{n-1}},$$

which is the required result.

**Part (iii)**

Combine the three sums on the left of (ii) into one. Its $r$th term is $(F_{r+1} - F_r - F_{r-1})/10^{r-1} = 0$, so the left-hand side is zero. With $F_0 = 0$ and $F_1 = 1$, this gives

$$89S_n = 10 - \frac{F_n}{10^n} - \frac{F_{n+1}}{10^{n-1}}. \qquad (*)$$

**Ask yourself** How do we know the two leftover terms vanish as $n \to \infty$?

This is where part (i) earns its keep. Taking $n = 1$ there gives $F_r \le 2^{r-1}$ for all $r \ge 1$, so

$$0 \le \frac{F_n}{10^n} \le \frac{1}{2}\Big(\frac{1}{5}\Big)^n, \qquad 0 \le \frac{F_{n+1}}{10^{n-1}} \le \frac{2^n}{10^{n-1}} = 10\Big(\frac{1}{5}\Big)^n,$$

and both bounds tend to 0. Letting $n \to \infty$ in $(*)$, we get $89S_n \to 10$, that is,

$$\sum_{r=1}^{\infty} \frac{F_r}{10^r} = \frac{10}{89}.$$

For the tail, apply part (i) with $n = 7$ (note $F_7 = 13$): for $r \ge 7$ we have $F_r \le 2^{r-7} \cdot 13$, so

$$\sum_{r=7}^{\infty} \frac{F_r}{10^r} \le \frac{13}{10^7} \sum_{j=0}^{\infty} \frac{2^j}{10^j} = \frac{13}{10^7} \cdot \frac{1}{1 - \frac{1}{5}} = \frac{65}{4} \times 10^{-7} = 1.625 \times 10^{-6} < 2 \times 10^{-6}.$$

*Digits of* $\frac{1}{89}$. The first six terms add up to

$$S_6 = 0.1 + 0.01 + 0.002 + 0.0003 + 0.00005 + 0.000008 = 0.112358.$$

The tail $T = \frac{10}{89} - S_6$ is strictly positive (its terms are positive) and less than $2 \times 10^{-6}$, so

$$0.112358 < \frac{10}{89} < 0.112360, \qquad \text{hence} \qquad 0.0112358 < \frac{1}{89} < 0.0112360.$$

Every number strictly between these bounds begins $0.011235\ldots$, so the first six digits after the decimal point of $\frac{1}{89}$ are $0, 1, 1, 2, 3, 5$. The strict lower bound matters: it rules out the expansion being $0.0112357\ldots$.

**Part (iv)**

Read the target in pairs of digits after the leading 00:

$$0.00\ 01\ 01\ 02\ 03\ 05\ 08\ 13\ 21\ 34\ 55\ldots$$

These pairs are $F_1, F_2, \ldots, F_{10}$, each written as two digits. So the number should be $\frac{1}{100}U$, where

$$U = \sum_{r=1}^{\infty} \frac{F_r}{100^r}.$$

*Summing $U$.* The series converges absolutely, because $F_r \leq 2^{r-1}$ by part (i). We can therefore shift and combine convergent series term by term, just as in (ii):

$$U - \frac{U}{100} - \frac{U}{100^2} = \frac{F_1}{100} + \sum_{r=2}^{\infty} \frac{F_r - F_{r-1} - F_{r-2}}{100^r} = \frac{1}{100}.$$

Here the $r = 1$ term is $F_1/100$ (since $F_0 = 0$), and every later bracket is zero. So $U \cdot \frac{9899}{10000} = \frac{1}{100}$, giving $U = \frac{100}{9899}$. The candidate is therefore

$$\frac{U}{100} = \frac{1}{9899},$$

with $r = 1$ and $s = 9899$, both positive integers less than 10000.

*Checking the digits.* Let $U_{10}$ be the sum of the first ten terms. Because each of $F_1, \ldots, F_{10}$ is less than 100, the terms do not overlap, and

$$U_{10} = 0.01010203050813213455$$

exactly, with 20 decimal places. By part (i) with $n = 11$ (and $F_{11} = 89$), the remaining terms satisfy

$$0 < U - U_{10} \leq \frac{89}{100^{11}} \sum_{j=0}^{\infty} \Big(\frac{2}{100}\Big)^j = \frac{89}{10^{22}} \cdot \frac{100}{98} < 10^{-20}.$$

So $U_{10} < U < U_{10} + 10^{-20}$, and the first 20 decimal digits of $U$ are those of $U_{10}$. Dividing by 100 shifts these digits two places to the right:

$$\frac{1}{9899} = 0.0001010203050813213455\ldots,$$

which is exactly the required start.

## Checks

- Long division gives $1/89 = 0.011235955\ldots$. The digits 011235 agree, and the next digits $9, 5, 5$ show the Fibonacci terms "carrying" into each other once they reach two digits.
- The tail beyond $r = 6$ in (iii) is actually $10/89 - 0.112358 \approx 1.55 \times 10^{-6}$, safely below our bound of $1.625 \times 10^{-6}$.
- A computer check of $1/9899$ to 22 places gives $0.0001010203050813213455$. The next pair is 90, not $F_{11} = 89$, because $F_{12} = 144$ overflows its two-digit slot and carries 1 into the one before.

## Common slips

- Getting the boundary terms wrong when re-indexing in (ii). Writing out the first and last terms of each shifted sum avoids this.
- In (iii), not showing that $F_n/10^n$ and $F_{n+1}/10^{n-1}$ tend to zero. Part (i) is there precisely to do this.
- Forgetting that the tail is *strictly* positive. An upper bound alone does not fix the sixth digit.

## Beyond STEP

The function $G(x) = \sum F_r x^r$ is the *generating function* of the Fibonacci numbers, and the shifting argument shows that $G(x) = x/(1 - x - x^2)$ for $|x| < \frac{1}{2}$. Putting in $x = \frac{1}{10}$ and $x = \frac{1}{100}$ gives $\frac{10}{89}$ and $\frac{100}{9899}$. Generating functions turn recurrences into algebra and are a central tool in combinatorics. Herbert Wilf's *generatingfunctionology* is a lively introduction. Making "rearranging convergent series" rigorous is a first-year analysis topic: see Abbott, *Understanding Analysis*.

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